Benthic Habitat Segmentation

1. Introduction

This report presents a deep learning pipeline for the semantic segmentation of benthic habitats using WorldView-3 8-band multispectral satellite imagery at Guam. The model classifies pixels into seven classes: Background, Seagrass, Coral, Macroalgae, Sand, Land, and Ocean across two study regions — Agana Bay and Manell-Geus.

2. Methodology

2.1 Data Preprocessing

Variable image sizes (528×447 to 1370×2051 px) were handled by tiling each image into 64×64 patches with a 48-pixel stride. Patches smaller than 64×64 at image edges were zero-padded. Per-band normalization used 2nd–98th percentile clipping followed by [0,1] scaling, which is robust to spectral outliers from sun glint and cloud shadows common in multispectral satellite imagery.

Class imbalance was addressed using inverse-frequency class weights in the loss function. Land (37.9%) and Ocean (27.4%) dominate the dataset, while Seagrass (3.1%) and Background (3.5%) are rare. The weights ensure the model penalizes errors on rare classes more heavily.

2.2 Model Architecture

A lightweight U-Net was used for pixel-wise segmentation. The encoder-decoder architecture with skip connections was chosen because it preserves spatial detail lost during downsampling — critical for thin habitat boundaries such as coral reef edges. The 8-channel input replaces standard RGB with all multispectral bands, including RedEdge, NIR1, and NIR2, which contain spectral signatures essential for distinguishing submerged vegetation from sand and coral.

Component	Channels	Notes
Input	8	8-band multispectral
Encoder 1	8 → 16	DoubleConv + BN + ReLU
Encoder 2	16 → 32	MaxPool + DoubleConv
Bottleneck	32 → 64	MaxPool + DoubleConv
Decoder 2	64+32 → 32	Upsample + skip + DoubleConv
Decoder 1	32+16 → 16	Upsample + skip + DoubleConv
Output	16 → 7	1×1 Conv, raw logits

Table 1: U-Net architecture overview (119,415 total parameters)

2.3 Loss Function and Hyperparameters

Weighted Cross-Entropy loss was used with inverse-frequency class weights. This directly addresses class imbalance by assigning higher loss penalties to rare classes, such as Seagrass (weight 0.32), compared to dominant classes like Land (weight 0.03).

Parameter	Value	Justification
Optimizer	AdamW	Stable convergence on small datasets
Learning rate	1e-4	Standard starting LR for U-Net
Scheduler	CosineAnnealingLR	Smooth decay, avoids sharp LR drops
Epochs	20	Convergence within the compute budget
Batch size	2	CPU memory constraint
Patch size	64×64	CPU memory constraint
Train patches	300	Subsampled for CPU training speed

Table 2: Training hyperparameters

3. Results

3.1 Training Curves

Training loss decreased steadily from 1.93 to 1.49 over 20 epochs. Validation loss showed high variance due to the small validation set (60 patches) but trended downward, reaching a best of 1.30 at epoch 12.

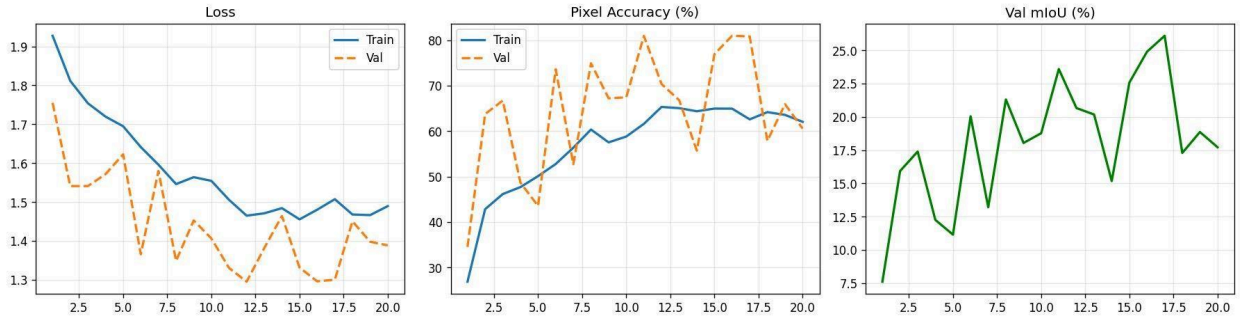


Figure 1: Training and validation loss, pixel accuracy, and mIoU over 20 epochs

3.2 Per-Image Evaluation

The model was evaluated on all 9 labeled training images using full-image sliding-window inference. Mean pixel accuracy was 62.0% and mean IoU was 34.5% across all images.

Image	Region	Pixel Acc.	mIoU
WV02052023.tif	Agana Bay	46.6%	28.8%
WV02202022.tif	Manell-Geus	74.0%	32.1%
WV220101215.tif	Agana Bay	42.1%	27.4%
WV220130115.tif	Agana Bay	57.3%	28.6%
WV220140205.tif	Agana Bay	81.2%	48.6%
WV220171020.tif	Agana Bay	40.8%	23.6%
WV220230219.tif	Agana Bay	77.2%	48.1%
WV320141230.tif	Manell-Geus	72.8%	38.6%
WV_01132022.tif	Manell-Geus	65.8%	34.5%
Mean	—	62.0%	34.5%

Table 3: Per-image evaluation on labeled data

3.3 Prediction vs. Ground Truth

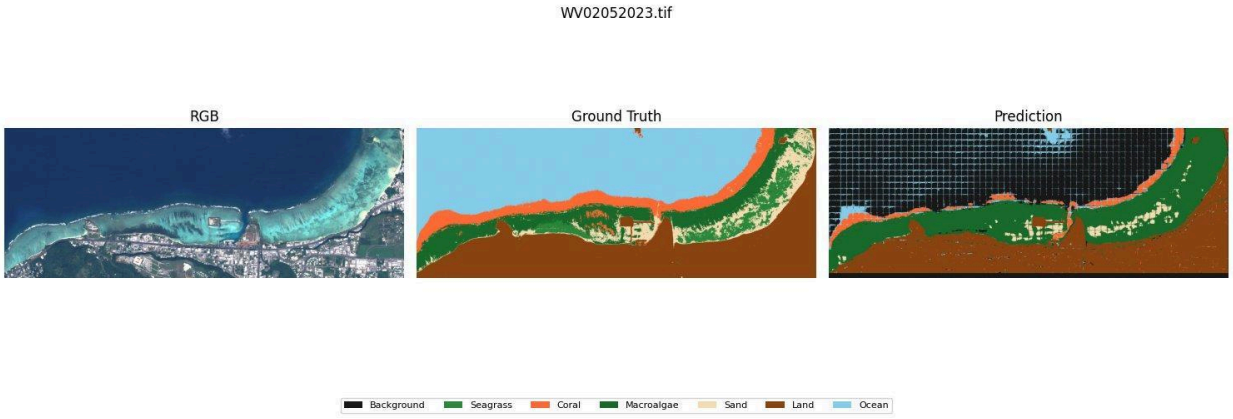


Figure 2: WV02052023 — RGB, ground truth, and prediction. The grid artifact in the ocean region is due to patch boundary effects, where the model predicts Background for deep water patches.

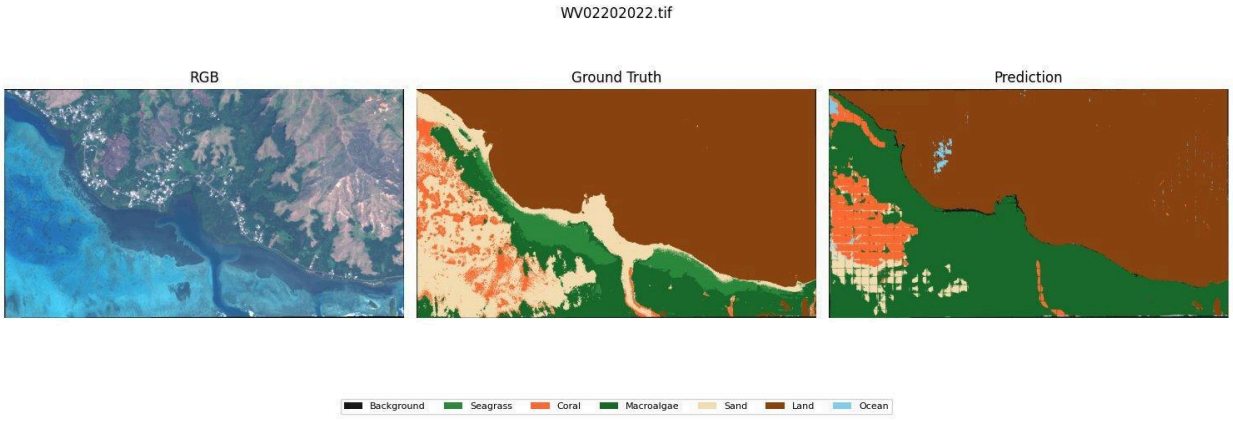


Figure 3: WV02202022 — RGB, ground truth, and prediction. Land and Macroalgae boundaries are well captured; Sand is partially confused with Macroalgae.

3.4 Confusion Matrix

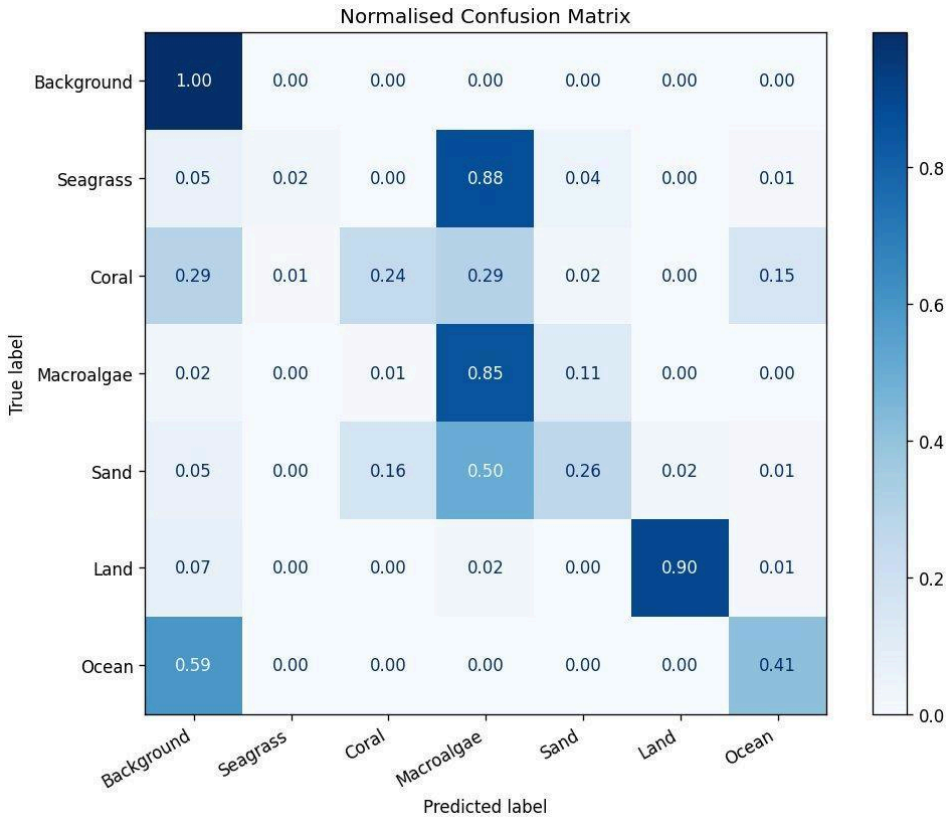


Figure 4: Normalised confusion matrix. Rows = true class, columns = predicted class.

Key observations from the confusion matrix: Land achieves 90% recall, confirming the model learns dominant classes well. Macroalgae has 85% recall. Seagrass is almost entirely misclassified as Macroalgae (88%) due to spectral similarity and extreme class imbalance. Coral is split between Background (29%), Macroalgae (29%), and correct predictions (24%). Ocean is frequently misclassified as Background (59%), likely because deep water patches outside the reef are assigned to the background class.

3.5 Error Analysis

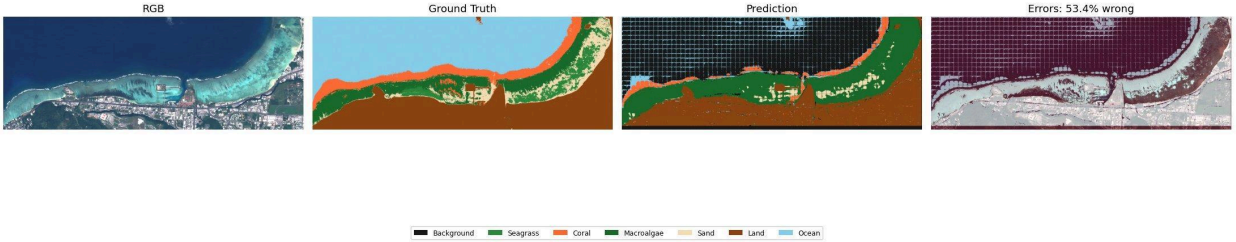


Figure 5: Error map for WV02052023. Red pixels indicate incorrect predictions (53.4% error rate). Errors concentrate in the ocean region (patch boundary artifacts) and along class boundaries between Coral, Sand, and Macroalgae.

3.6 Test Set Performance

The model was evaluated on 21 unlabeled test images (15 Agana Bay, 6 Manell-Geus). Where ground truth annotations were available, mean accuracy was 17.4% for Agana and 26.1% for Manell-Geus. The large drop from training performance is expected due to temporal domain shift — test images span 2010–2024 while training data is more limited, meaning different water clarity, turbidity, and sun angle conditions affect spectral signatures significantly.

Region	Images	Mean Acc.	Mean mIoU
Agana Bay	15	17.4%	3.3%
Manell-Geus	6	26.1%	9.9%

Table 4: Test set performance

4. Discussion

4.1 What Worked

The U-Net architecture successfully learned spatial structure for dominant classes. Land (90% recall) and Macroalgae (85% recall) are consistently well-segmented across both regions. The inverse-frequency weighting reduced the tendency to predict only dominant classes and helped the model learn minority class boundaries.

4.2 Sources of Error

Seagrass performance was near zero throughout training. With only 3.1% of pixels and spectral overlap with Macroalgae, 300 training patches and 20 epochs are insufficient to learn its signature. Ocean background confusion arises because deep water outside reef areas shares similar spectral profiles with the background class. The grid artifact visible in predictions is a patch boundary effect — predictions at patch edges are less reliable because the model sees zero-padded context.

The large train-to-test performance drop reflects a temporal domain shift. Images from different years and seasons exhibit different atmospheric conditions, water turbidity, and algal cover, making generalization difficult without domain adaptation.

4.3 Potential Improvements

Longer training with more patches, a larger model on a GPU, and Dice loss would improve minority class recall. Test-time augmentation (averaging predictions from flipped/rotated versions of each patch) would reduce boundary artifacts. Data augmentation with spectral jitter could help the model generalize across acquisition dates.

5. Conclusion

A lightweight U-Net was trained for benthic habitat segmentation on 8-band WorldView-3 imagery. The model achieved 62% pixel accuracy and 34.5% mIoU on labeled training data, with strong performance on Land and Macroalgae but limited success on ecologically rare classes such as Seagrass and Coral. All predicted masks for test images have been submitted in predicted_masks.zip.